

11a. Osmoreguation

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Pre-class materials

! Let's finish strong!

- We have **only 3** class sessions left to finish all of our content. After that, we have final student presentations.
- **Attendance is mandatory** at all remaining classes.
- **Come prepared by reading the book**, and our discussions can **probe deeper**. This will make your **homeworks, labs, and papers** so much easier.
- **Remember that effort counts**, and will be taken into account in the final grades. If your grade is on the borderline, effort will translate into a “bump up”. You show effort in many ways, including attendance and sincere participation.

Announcements/Reminders

- Welcome back from Thanksgiving!
- Due Friday Design 3 [[turn in](#)]
- Homework 7 Due Monday on Lamaku – [[schedule](#)]
- Last lab this week! **Prepare for the Human Renal Lab starting from the night before lab.** [[lab 12](#)]
- Design 4 is posted [[suggestions](#)] or feel free to pursue your own interests and design your own project! Due the week of finals.

Week 15 Discussion Groups

Group	Partner 1	Partner 2	Partner 3
1	Dallas	Emma	Peigi

Group	Partner 1	Partner 2	Partner 3
2	Kyra	Niki	Reagan
3	Alice	Carla	Paris
4	Darwin	Manuel	Trey

Wednesday

- **Reading assignment:** Withers skim beginning Ch. 16 for definitions in list below, also read Water & Ion Budgets pp. 788-790, Vertebrates pp. 798-803, Terrestrial Environments pp. 806-812, Terr. Verts. 822-827. Withers is actually very readable on this topic, and some of it will be familiar to you already:).
- [Osmoregulation discussion] [slide deck]
- definitions:
 - solutes,
 - osmosis,
 - osmolarity/osmolality,
 - ionoregulation,
 - osmoregulation/osmoconforming,
 - compatible solutes,
 - perturbing solutes,
 - counteracting solutes,
 - osmotic permeability (& formulae)
- The components of **Water Budgets** (input/output) as well as **Ion Budgets**
- The water and ion challenges of **freshwater**, **marine**, and **terrestrial** environments

Osmoregulation Podcast

<https://youtu.be/1vQxLFoPCmM>

- If there is time, start [Excretion questions 1,2] [slide deck 2]
- Be prepared to diagram and discuss the routes of water and ion inflow and outflow for terrestrial animals (human), freshwater fish (q1), marine fish (q2), and marine elasmobranchs (q3).
 - This time, start with the osmotic gradient facing the animal. Clearly label the mOsm of the media outside, as well as mOsm inside (use the extracellular fluid as the inside compartment).
 - Consider the gradient facing the skin :) also.
 - Label special organs or sites of major exchange.

- Indicate **water** and **ions** separately, and whether it is active or passive movement, and for ions - which ones (details matter).
- **Look in your book!**
- For excretion think about nitrogenous waste (osmoreg question 5) and what mechanisms organisms have for their excretion (excretion questions 1,2)

Table 2: Organs involved in osmoregulation

Group					
Fish	Kidneys	Gills	Bladder	Intestines	
Amphibians	Kidneys	Gills	Bladder	Intestines	Skin
Reptiles	Kidneys			Intestines	(Salt Glands)
Birds	Kidneys			Intestines	(Salt Glands)
Mammals	Kidneys				

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